Title:
Evaluation of a Zeolite-Derived Aerosol as a Cloud Condensation Nuclei (CCN) Analog

Author:
Aaron Corbin Mazie
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1. Purpose

This brief proposes a low-risk, laboratory-scale evaluation of a modified zeolite-based aerosol as a potential cloud condensation nuclei (CCN) analog. The objective is to determine whether a porous aluminosilicate material, following cation exchange and surface modification, exhibits measurable CCN activation under controlled supersaturation conditions.

No operational, weather modification, or field deployment claims are made. This is strictly a microphysical characterization exercise.

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2. Background

Cloud condensation nuclei play a central role in droplet activation and cloud microphysics, commonly parameterized via hygroscopicity (κ) within the framework of Köhler theory.

Common CCN proxies (e.g., ammonium sulfate, sodium chloride) are highly soluble but lack structural complexity. Porous materials such as zeolites offer:

- High internal surface area
- Tunable cation composition
- Potential for water adsorption and retention

This raises the question of whether engineered zeolite particles can function as moderate-hygroscopicity CCN analogs within atmospherically relevant supersaturation ranges.

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3. Material Description

Base Material:
Commercial Zeolite 13X (FAU-type framework)

Modifications:

- Cation exchange: Partial substitution of Na^+ with Ca^{2+} or Mg^{2+}
- Surface loading: Low-level impregnation with NaCl or K_2SO_4
- Target particle size: 50–200 nm (post-milling/classification)

Expected Properties:
- Hygroscopicity parameter $\kappa \approx 0.1$–0.3
- Moderate water uptake at RH 70–90%
- Non-reactive, non-toxic composition

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4. Proposed Experimental Protocol

The following procedure is designed to align with standard CCN measurement capabilities already in use.

4.1 Aerosol Generation
- Disperse dry or nebulized particles into a controlled chamber
- Target concentration: 10^2–10^4 particles/cm^3

4.2 Size Selection
- Use SMPS or equivalent to isolate particle diameters:
  - 50 nm
  - 100 nm
  - 150 nm

4.3 CCN Activation Measurement
- Instrument: CCN counter (continuous-flow thermal gradient type)
- Supersaturation scan range: 0.1% → 1.0%

Output:
- Activation fraction vs supersaturation
- Critical supersaturation (Sc) per particle size

4.4 Baseline Comparison
- Ammonium sulfate (reference standard)
- Sodium chloride (secondary benchmark)

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5. Expected Outcomes

If the material behaves as hypothesized:

- Detectable CCN activation within 0.2–0.8% supersaturation
- Size-dependent activation consistent with κ-framework predictions
- Lower activation efficiency than fully soluble salts, but within measurable range

A null result (no activation) would still provide useful constraints on the role of porosity vs solubility in CCN behavior.

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6. Significance

This test would:

- Evaluate a structurally distinct CCN analog
- Contribute to understanding of porous aerosol behavior in cloud activation
- Provide comparative data relative to conventional soluble particles

The experiment is compatible with existing aerosol and cloud microphysics infrastructure and does not require new instrumentation or field deployment.

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7. Safety and Environmental Considerations

- Zeolite and associated salts are widely used and non-toxic in laboratory contexts
- Standard nanoparticle handling precautions apply (PPE, ventilation)
- No reactive or hazardous chemistries are involved

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8. Summary

This proposal outlines a minimal-effort, instrument-compatible test of a modified zeolite aerosol for CCN activity. The required materials are commercially accessible, and the experimental protocol aligns with established measurement techniques.

The study is intended solely to assess activation behavior under controlled conditions and does not imply any application beyond laboratory-scale aerosol research.

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